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Server Sensor Update Behavior through an Aggregator

In many deployments, Windows Server Sensors and Linux Server Sensors do not communicate directly with the Data Processor (DP). Instead, they send and receive data through a Modular Sensor configured as an aggregator. The aggregator serves as an intermediary between the Server Sensors and the DP, forwarding data and retrieving update files when sensors request them.

Some deployments use a single Modular Sensor as an aggregator, and some use two Modular Sensors configured as a high availability (HA) pair. The use of an HA aggregator pair does not change the update or upgrade behaviors described here. Windows and Linux Server Sensors continue to retrieve updates and upgrades according to their operating system.

Each Server Sensor independently runs a scheduler, which periodically checks the DP for updates. The scheduler operates on the Server Sensor itself, not on the aggregator or DP, and initiates update requests. When an update is available, the sensor downloads the necessary files through the aggregator. If the download fails, the sensor might attempt to retrieve the update directly from the DP as a fallback.

In this section, you can learn how Server Sensors retrieve updates when an aggregator is in use, what types of files are downloaded, and under what conditions a sensor might bypass the aggregator to communicate directly with the DP.

Scheduled Updates through an Aggregator

When a Windows Server Sensor or Linux Server Sensor is deployed with an aggregator, the update process follows four main steps:

1. A Server Sensor initiates a request to check for updates.

   Every 30 minutes (by default), the scheduler runs on the Server Sensor and checks the DP for available updates.

   - **Download Operations**: The Server Sensor attempts to retrieve updates by first contacting the primary aggregator. If that fails, it attempts to contact the secondary aggregator, and finally the Data Processor itself.

- **Upgrade Operations – Windows Server Sensor**: Attempts the primary aggregator, then the secondary aggregator, and finally the DP.
- **Upgrade Operations – Linux Server Sensor**: Uses the configured aggregator only. No fallback to direct DP connection.

Operation Type	Sensor Type	Behavior
Download	Windows or Linux	Attempts primary aggregator → secondary aggregator → DP
Upgrade	Windows	Attempts primary aggregator → secondary aggregator → DP
Upgrade	Linux	Uses aggregator only

2. The DP responds through the aggregator to the sensor with update tasks.

   The DP provides information on available updates, including new software versions, configuration files, and security rule updates.

3. The Server Sensor downloads update files through the aggregator.

   Files are retrieved from the DP through the aggregator. The downloaded files are stored in the `C:\ProgramData\StellarCyber\Windows Agent Sensor\config` and `C:\ProgramData\StellarCyber\Windows Agent Sensor\aella_update\tmp` directories on Windows and the `/etc/stellarcyber/` directory on Linux.

4. The update process is logged in `aella_update.log`.

   The sensor logs all download attempts and update executions. If the sensor fails to download through the aggregator, it logs the fallback attempt to direct DP communication.

   Example log entries:

   ```
   2025-04-15 12:14:03,316|aella_update|INFO|Downloaded from aggregator 192.168.5.5
   ```

   ```
   2025-04-15 12:14:04,249|aella_update|INFO|Fallback to direct download from cm-acmecorp.stellarcyber.cloud
   ```

Downloaded Update Files

During an update, a Server Sensor might download the following files:

Files	Purpose
`ds_tasks.json`	Checks for pending update tasks on the DP.
`.msi`/`.msp` files	Update packages for upgrading to a new software version (Windows only). Also included is the `download_image.exe` file, which is extracted after download and used to retrieve updates from the DP.
Custom parsers	Files that update the ability of the sensor to process and normalize logs.
Threat intelligence updates	Updated detection rules and heuristics for security analysis.

Fallback to Direct DP Communication

In certain situations, a Server Sensor might attempt to bypass the aggregator and connect directly to the DP over TCP port 8443 to download updates. This is a fallback mechanism triggered when the sensor can't retrieve files through the aggregator.

Reasons to Fall Back

- The sensor encounters an aggregator communication failure.

  If the sensor fails to download update files through the aggregator, it retries directly from the DP. The failure can be caused by network issues, misconfigured firewall rules, or connectivity problems between the aggregator and DP.

- TCP port 8443 is not open from the aggregator to the DP.

  The aggregator must have TCP port 8443 open to allow update file transfers from the DP. If this port is blocked, the sensor cannot download files through the aggregator and must fall back to direct DP communication.

- The aggregator is temporarily unavailable.

  If the aggregator is down or undergoing maintenance, the sensor switches to direct communication with the DP.

- The sensor experiences a failed initial download attempt.

  If the sensor attempts to download `ds_tasks.json` or an update package and encounters an error, it automatically retries using the direct IP address of the DP.

Update Logging Behavior

The sensor records its update attempts in the `aella_update.log`, including whether the file was successfully downloaded via the aggregator or if it switched to direct DP communication.

Example log entries:

```
2024-08-06
12:14:03,316|aella_update|INFO|"c:\ProgramData\StellarCyber\aella_updat
e\20231002_4.3.7_12c2f39\download_image.exe" -cm 192.168.5.5 -path
"C:/ProgramData/StellarCyber/Windows Agent Sensor/config" -image
ds_tasks.json -flag json_task

2024-08-06
12:14:04,249|aella_update|INFO|"c:\ProgramData\StellarCyber\aella_updat
e\20231002_4.3.7_12c2f39\download_image.exe" -cm cm-
acmecorp.stellarcyber.cloud -path "C:/ProgramData/StellarCyber/Windows
Agent Sensor/config" -image ds_tasks.json -flag json_task
```

In this example, the first attempt downloads `ds_tasks.json` from the DP via the aggregator at 192.168.5.5. When the aggregator download fails, the sensor tries to download the file directly from the DP at `cm-acmecorp.stellarcyber.cloud`.

Preventing Direct DP Access

To ensure that all update downloads occur through the aggregator and avoid unexpected direct connections to the DP, take the following steps:

1. Verify that TCP port 8443 is open between the aggregator and DP.

   This allows the aggregator to retrieve update files from the DP successfully.

2. Check firewall rules and routing configurations.

   Ensure that traffic between the Server Sensor and aggregator is not being blocked or disrupted.

3. Monitor the logs for repeated fallback behavior.

   Regularly check aella_update.log for signs that the sensor is bypassing the aggregator.

4. Ensure the aggregator is online and functioning.

   If the aggregator is temporarily unavailable, sensors default to direct DP communication.

Summary

Windows and Linux Server Sensors configured to use an aggregator should normally download updates through it. However, if the aggregator becomes unavailable or encounters connectivity issues, Windows Server Sensors fall back to direct communication with the Data Processor. Linux Server Sensors, by contrast, always rely on the configured aggregator for updates and upgrades even if connectivity issues occur. All fallback behaviors are recorded in `aella_update.log`.

Keeping TCP port 8443 open, monitoring logs for unexpected behavior, and maintaining proper firewall and network configurations ensures consistent and reliable update and upgrade operations. For troubleshooting, refer to **System | Logs** in the Stellar Cyber UI or examine `aella_update.log` on the Server Sensor host.

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